

Estimation of Quantum Fisher Information by variational quantum circuits on a quantum computer

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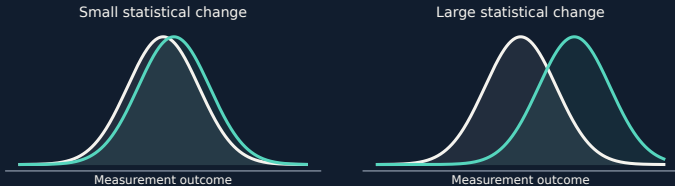
In collaboration with the Italian Space Agency (ASI)

Project: *Quantum Computers for Space Exploration*



Fisher Information

Local sensitivity quantifies the distinguishability of nearby parameter values.



Fisher information

Sensitivity of an observed probability distribution to changes in an unknown parameter

Quantum Fisher information (QFI)

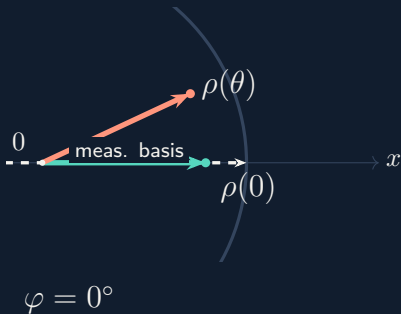
Intrinsic sensitivity of a quantum state to a parameter, independent of the measurement basis chosen.

Illustrative distributions. QFI is a local benchmark; attainable precision also depends on the measurement and estimation assumptions.

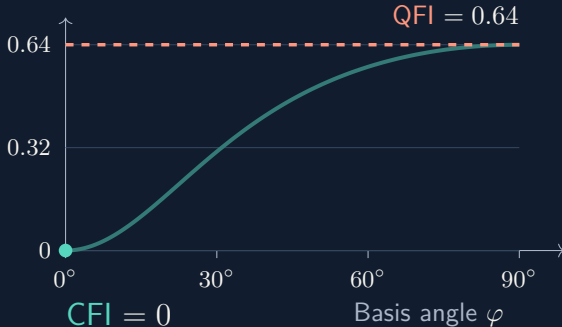
Fisher Information and Measurements

Two states inside the Bloch sphere; the dashed measurement axis rotates.

Two mixed states



Accessible information



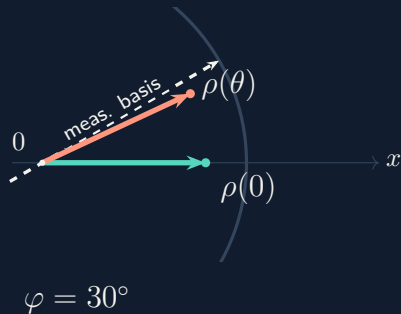
The measurement changes CFI, the QFI remains fixed.

Mixed states, $r = 0.8$; finite separation shown for clarity. CFI is local at $\theta = 0$ (radians), not a two-state finite difference. Appendix J.

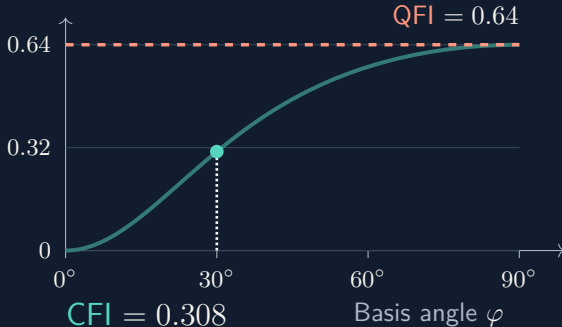
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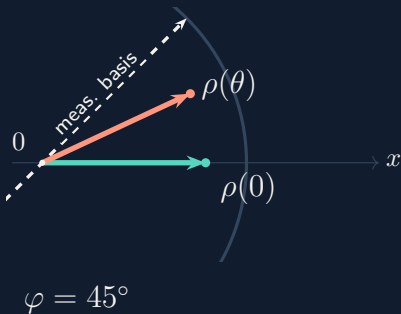
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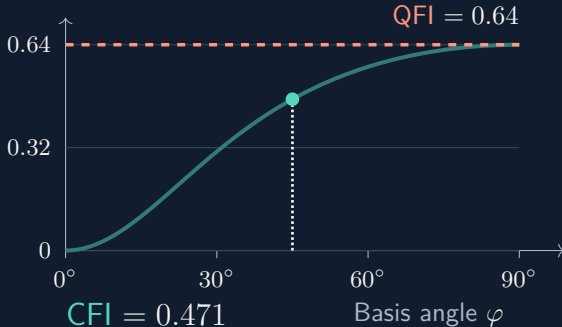
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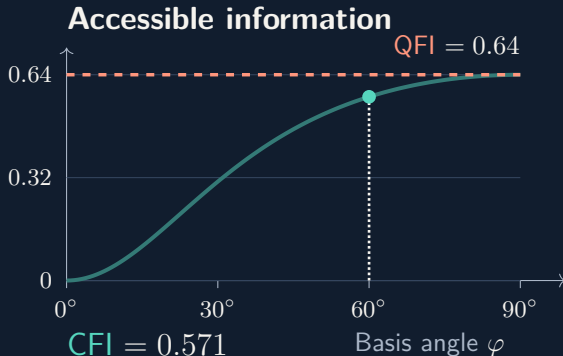
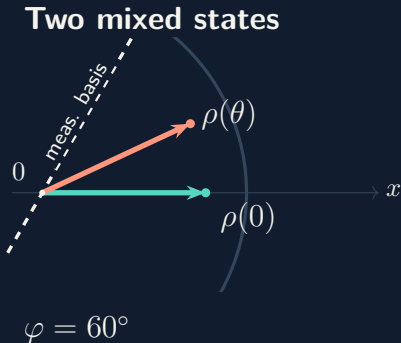


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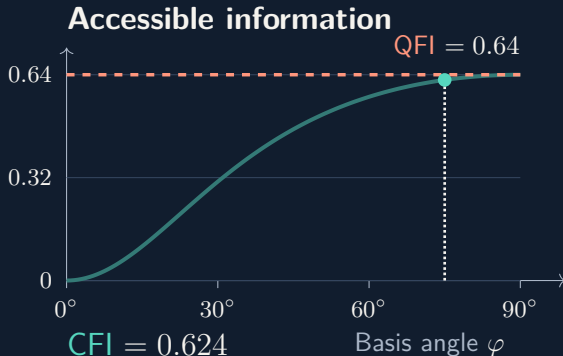
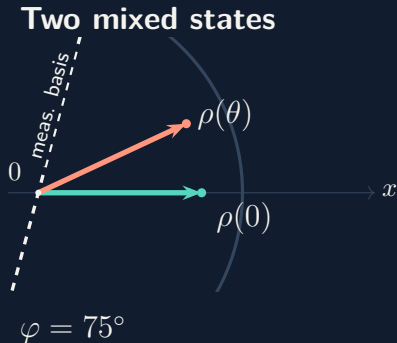


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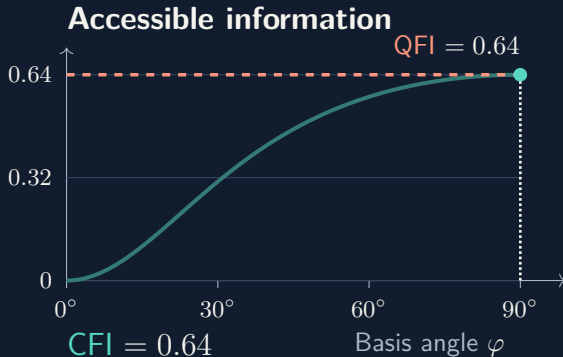
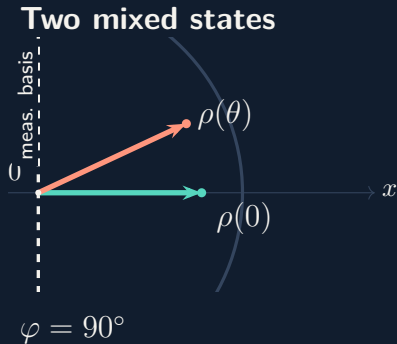


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Learning the dominant components of a mixed state

Spectral truncation provides a compact representation for sensitivity estimation.

We retain the dominant eigencomponents to represent mixed states efficiently.

$$\rho_m = \sum_{i=1}^m \lambda_i |v_i\rangle\langle v_i|$$

Retain the m largest eigenvalues and their associated eigenvectors.

VQSE — Variational Quantum State Eigensolver

Learns the leading eigenvalues and a circuit that prepares their eigenvectors.

PCA — Principal Component Analysis: an analogy

Both retain dominant spectral components: state probabilities here, variances in PCA.

TQFI — Truncated Quantum Fisher Information

The retained components provide input to fidelity-based QFI bounds.

VQSE / VQFIE: Cerezo et al. (2022); Beckey et al. (2022). Truncation is subnormalized, not renormalized. Finite-step scope: Appendix A.

Applications

Study QFI and TQFI through numerical simulation and quantum-circuits.

Compare exact numerical benchmarks with variational spectral estimation.

Quantum magnetometry

Estimate a magnetic field with an interacting spin system.

Preparation, restricted access and spectral truncation.

Exoplanet imaging

Resolve a faint companion with spatial-mode measurements.

Optical channels, accessible information and experimental requirements.

Connect information in the quantum state to information accessible in measurements.

Magnetometry: numerical and circuit methods. Optical application: channel analysis and proposed variational protocol; experimental TQFI was not obtained.

Parallel numerical and quantum-circuit workflows

NUMERICAL SIMULATION

Preparation

Hamiltonian
Time evolution

Access

→ Partial trace

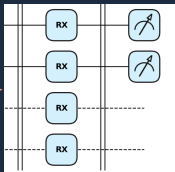
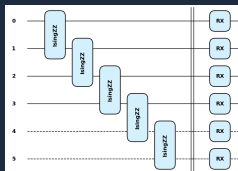
Spectrum

→ Exact spectral
decomposition

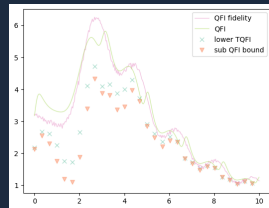
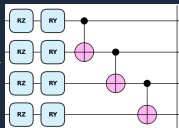
Information

Exact SLD QFI
TQFI bounds

QUANTUM CIRCUIT



VQSE



Schematic circuit layouts and example output. Implementations can be exchanged at matched interfaces; physical stage order is preserved.

APPLICATION I

Quantum Magnetometry

Toy Model: a noisy Quantum Magnetometer

Encode a magnetic field, restrict access, study QFI



Periodic spin chain
4 retained / 2 discarded

Ground state	Lowest-energy preparation
Depolarized	Fixed depolarizing noise
Thermal	Field-dependent populations

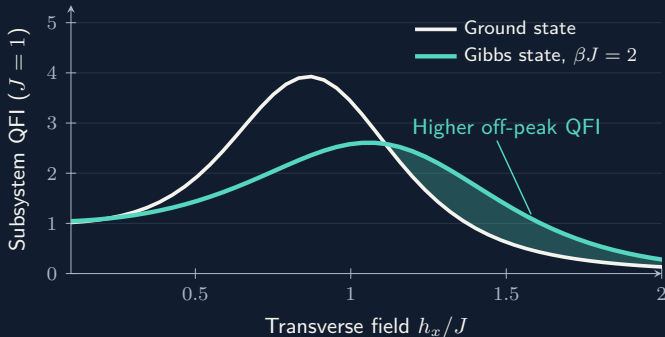
We studied:	 Local QFI retained
	 Effects of preparation
	 Noise, temperature effects

Tune temperature for peak sensitivity or a broader useful field range.

Transverse-field Ising model (TFIM), $N = 6$, $n = 4$, periodic boundaries, $J = 1$. Thesis: NoiseAnalysis, Project 9.

Temperature broadens the sensing range

At $\beta J = 2$, Gibbs preparation retains more information beyond the ground-state peak.



Broader sensitivity

More QFI beyond the ground-state peak.

Uncertain field

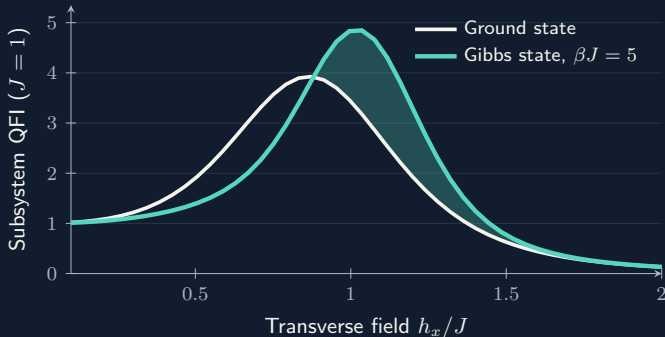
Useful across a wider field range.

When the field is uncertain, off-peak sensitivity matters as well as the maximum.

Exact SLD QFI; periodic $N = 6$, $n = 4$, $J = 1$. Shading: Gibbs QFI above ground-state QFI. Different preparation families.

Cooling strengthens the sensing peak

At $\beta J = 5$, the Gibbs state exceeds the ground-state peak in the same spin subsystem.



Stronger peak

Higher QFI around the thermal optimum.

Known field range

Useful near the target field.

When the field lies near the optimum, cooling can provide greater local sensitivity.

Exact SLD QFI; periodic $N = 6$, $n = 4$, $J = 1$. Shading: Gibbs QFI above ground-state QFI. Different preparation families.

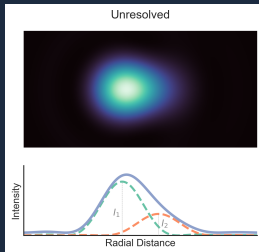
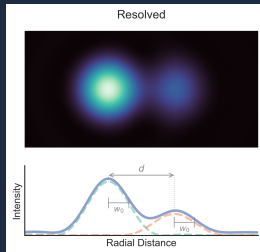
APPLICATION II

Exoplanet Detection

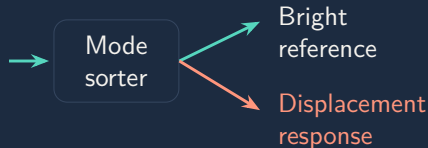
Spatial-mode measurements for exoplanet imaging

A faint companion must be resolved in the presence of a bright reference source.

Direct imaging



Spatial-mode sorting



Photon counts in a spatial-mode basis.

SPADE changes the measurement basis to reveal weak spatial signals.

Thesis illustrations; schematic mode sorter. SPADE: spatial-mode demultiplexing. Tsang et al. (2016); Santamaria et al. (2024).

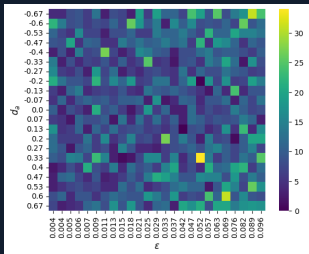
SPADE detector

Experimental observations and identifiability

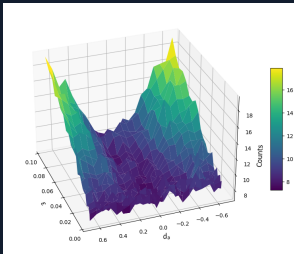
The completed ASI Matera experiment supplied aggregate first-order counts.

52,500 count observations: 525 settings $\times$ 100 repetitions; 10 ms each

ONE REPETITION



MEAN OF 100



Missing observables

Separate modal outputs

Other measurement bases

Incident flux and failure outcomes

Repeated counts constrain the mean response; they do not identify the quantum state.

Thesis: Exoplanet Experiment. d_a : displacement; ϵ : source ratio. Aggregate counts; no basis rotations or experimental TQFI.

Why describe the receiver with quantum channels?

The model includes mode populations and coherence before the measurement.

CLASSICAL PROBABILITY EMBEDDING

$$\mathbf{p} = \begin{pmatrix} p_0 \\ p_1 \end{pmatrix} \longmapsto \rho_{\text{diag}} = \begin{pmatrix} p_0 & 0 \\ 0 & p_1 \end{pmatrix}$$

Classical channel: $\mathbf{p}' = R\mathbf{p}$

Redistributes probabilities between outcomes.

QUANTUM COHERENCE

$$\rho = \begin{pmatrix} p_0 & c \\ c^* & p_1 \end{pmatrix}$$

Quantum channel: $\rho' = \mathcal{E}(\rho)$

Acts on populations and coherences.

Quantum channels describe optical effects before the detector produces classical counts.

Two-outcome illustration: $p_i \geq 0$, $p_0 + p_1 = 1$, $|c|^2 \leq p_0 p_1$. A state vector of probabilities does not determine a quantum channel.

From physical assumptions to predicted counts

A modular model connects source preparation, optical effects and detector readout.



Choose physically motivated channels and check their mathematical validity.

Physical channel choices

Kraus operators for loss, dephasing and coherent mode mixing.

Classical calibration

Fit throughput, background, offset and coupling from recorded counts.

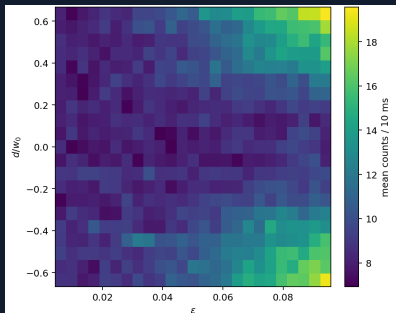
Simulate the chosen channel sequence and compare its output with the observations.

Thesis: provisional channel study. Optical assumptions are replaceable; the count record does not determine their individual values. Details: Appendices K–M.

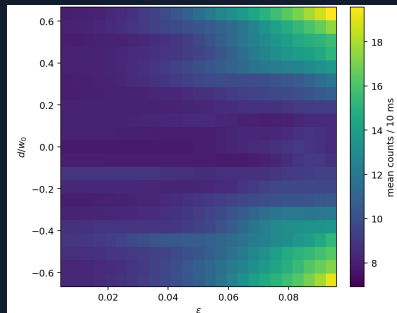
The model reproduces the observed mean response

Both maps show a central minimum and larger counts at greater separation and source ratio.

Observed setting means



Modelled setting means



The mean-count pattern is captured by the chosen channels with classical calibration.

Thesis appendix: provisional channel model, top panels. In-sample mean comparison; assumed optical channels are not individually identified by this agreement.

Main contributions

Simulation, preparation studies, data interpretation and a variational optical proposal.

- **Quantum and numerical sensing pipeline**

Simulated quantum sensors under noise and decoherence, estimating QFI and TQFI.

- **Preparation and noise models**

Compared ground and Gibbs states, depolarization and restricted access to the sensor.

- **Quantum-channel interpretation of SPADE data**

Modelled SPADE with quantum channels and fitted the observed mean counts.

- **VQ-SPADE: a proposed optical eigensolver**

A trainable mode rotation and SPADE readout would diagonalize the optical state.

Thesis: Methods, Noise Analysis, Exoplanet Experiment and Conclusions. VQ-SPADE is proposed for a future experiment.

Spectral truncation and the local-QFI limit

Dominant eigencomponents bound a fidelity quantity at finite parameter displacement.

$$\mathcal{I}_{\Delta} = \frac{8[1 - F(\rho_{\theta}, \rho_{\theta+\Delta})]}{\Delta^2}$$

F is root fidelity.

The retained eigenspace is state dependent; it is not a fixed detector-mode cutoff.

At regular operating points, $\mathcal{I}_{\Delta} \rightarrow \mathcal{F}_Q$ as $\Delta \rightarrow 0$.

Finite-step bounds need not be pointwise bounds on the exact SLD QFI.

Thesis: Methods; Mathematical Background. Cerezo et al. (2022); Beckey et al. (2022).

The ansatz metric and its scope

The physical sensing parameter and the trainable circuit controls define different derivatives.

$$F_{\mu\nu}^{\text{ans}} = 4 \operatorname{Re}[\langle \partial_\mu \psi | \partial_\nu \psi \rangle - \langle \partial_\mu \psi | \psi \rangle \langle \psi | \partial_\nu \psi \rangle]$$

Pure ansatz: $F^{\text{ans}} = 4g_{\text{FS}}$, the Fubini–Study metric up to convention.

$$\min_{\Delta\alpha} \nabla C^\top \Delta\alpha + \frac{1}{2\eta} \Delta\alpha^\top (F^{\text{ans}} + \lambda I) \Delta\alpha$$

Stationarity gives the regularized natural-gradient update.

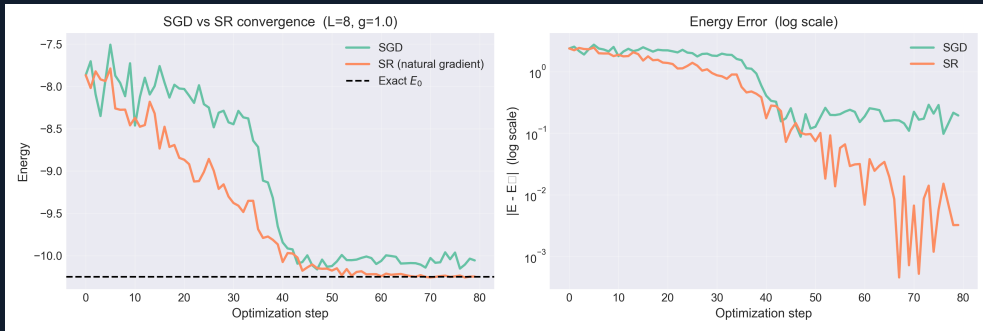
The quadratic term constrains state change, not objective curvature.

Mixed VQSE inputs require a mixed-state metric; a pure-state metric is otherwise a surrogate.

Thesis: Mathematical Background; VQSE. A full metric has quadratic parameter-count overhead; no general training speedup is claimed.

Stochastic reconfiguration in a neural quantum state

A separate pure-state RBM example illustrates metric-based optimization.



SR lowers the energy error in this run; it does not benchmark mixed-state VQSE.

Thesis figure: VQSE chapter. TFIM at $\hbar_x/J = 1$; 80 steps, 300 Monte Carlo samples/step, $\eta = 0.05$, SR $\lambda = 10^{-4}$; same seed.

Thermal sensitivity and data processing

The Gibbs and ground-state families need not have the same field response.

$$\rho_{\beta}(h_x) = \frac{e^{-\beta H(h_x)}}{\text{Tr } e^{-\beta H(h_x)}}$$

The field changes both eigenvectors and thermal populations.

Partial trace still obeys data processing within each family:

$$\mathcal{F}_Q[\text{Tr}_E \rho_{\beta}(h_x)] \leq \mathcal{F}_Q[\rho_{\beta}(h_x)]$$

For an uncertain field, choose an operating range or prior.

These curves do not establish a universal Bayesian or minimax optimum.

The comparison changes the preparation family; it does not reverse a fixed noise channel.

Thesis: Noise Analysis. Finite system; exact SLD QFI.

Proof of Kraus-unitary freedom

A change of Kraus basis leaves the channel unchanged.

$$\widetilde{K}_a = \sum_b u_{ab} K_b, \quad U^\dagger U = I$$

$$\sum_a u_{ab} u_{ac}^* = \delta_{bc}$$

Orthonormal columns of U .

$$\sum_a \widetilde{K}_a \rho \widetilde{K}_a^\dagger = \sum_{b,c} \delta_{bc} K_b \rho K_c^\dagger = \sum_b K_b \rho K_b^\dagger$$

Isometry in the Kraus index leaves the action on every input unchanged.

A physical realization adds assumptions about the environment; the reduced map alone does not select them.

Standard Kraus equivalence: an isometry, or a unitary after padding with zero operators. Algebra shown explicitly for this presentation.

A classical response does not fix a quantum channel

Embedding a stochastic map is a model construction, not process tomography.

$$\mathbf{p} \mapsto \text{diag}(\mathbf{p}), \quad \mathbf{p}' = R\mathbf{p}$$

For a column-stochastic R , one valid quantum extension is

$$K_{ji} = \sqrt{R_{ji}}|j\rangle\langle i|, \quad \mathcal{E}_R(\rho) = \sum_{ji} R_{ji} \rho_{ii} |j\rangle\langle j|$$

This extension measures and prepares in the chosen basis, erasing coherences.

For $R = I$, both the identity channel and complete dephasing reproduce the same populations.

Different channels can agree on the measured populations; this is distinct from Kraus freedom of one channel.

Thesis: classical/quantum embedding and HG coherence discussion. Channel-extension example derived for the presentation.

The loss map and its physical checks

The input mode space is embedded in an output space containing an orthogonal failure flag.

$$\mathcal{L}_\tau(\rho) = K_t \rho K_t^\dagger + \text{Tr}[(I - K_t^\dagger K_t)\rho] |\text{fail}\rangle \langle \text{fail}|$$

For any ancillary system, Kraus form preserves positivity:

$$(\mathcal{L} \otimes \text{id})(X) = \sum_a (K_a \otimes I) X (K_a^\dagger \otimes I) \geq 0 \text{ for } X \geq 0.$$

Boundary checks: $\tau_j = 1$ transmits mode j ; $\tau_j = 0$ sends it to failure.

Normalization: transmission plus failure has unit trace.

Physical validation: mode-injection measurements determine the τ_j .

Renormalizing the transmitted branch alone changes the question to postselected performance.

Thesis: mode-dependent loss and flagged mode reduction. The record lacks independent modal-transmission calibration.

Predictive calibration of the observed response

A physical mean with a Gaussian-process correction predicts held-out setting means.

$$\mu(d_a, \epsilon) = \mu_0(d_a, \epsilon) \exp[f(d_a, \epsilon) + \xi]$$

Grouped five-fold validation; all components refitted within each training fold.

Held-out structure	RMSE	Standardized SD	95% coverage
Checkerboard cells	0.606	1.004	93.52%
Distance rows	0.734	1.041	94.67%
Source-ratio columns	0.609	1.009	93.71%

The validated target is the mean of 100 repetitions; this does not identify a complete quantum channel.

Thesis: Exoplanet Experiment Appendix, hierarchical calibration validation. RMSE in counts per 10 ms.

From SPADE to a variational optical measurement

VQ-SPADE is a proposed optical realization of variational state eigensolving.

$$p_k(\boldsymbol{\alpha}; d_a) = \langle HG_k | V(\boldsymbol{\alpha}) \rho(d_a) V^\dagger(\boldsymbol{\alpha}) | HG_k \rangle$$

A programmable coherent mode transformation provides the trainable basis.

Resolved outputs, flux and failure events provide normalized probabilities.

Nearby states and additional overlap measurements support fidelity-based QFI bounds.

The present record motivates the acquisition design; it does not implement this protocol.

Thesis: proposed VQ-SPADE scheme. Additional independent channel and background calibrations are required.

Mixed states and measurement-dependent information

Two finite-separated vectors illustrate one state family; Fisher information is evaluated locally.

$$\rho(\theta) = \frac{I + r(\cos \theta X + \sin \theta Z)}{2}, \quad r = 0.8, \quad \rho \equiv \rho(0)$$

$$M_{\pm}^{(\varphi)} = \frac{I \pm (\cos \varphi X + \sin \varphi Z)}{2}, \quad p_{\pm}(\theta) = \frac{1 \pm r \cos(\theta - \varphi)}{2}$$

$$\mathcal{F}_C(\varphi)|_{\theta=0} = \frac{r^2 \sin^2 \varphi}{1 - r^2 \cos^2 \varphi} \leq r^2 = \mathcal{F}_Q$$

At $\varphi = 0^\circ$: CFI = 0. At 45° : CFI = $8/17 \simeq 0.471$.

At $\varphi = 90^\circ$, measuring Z attains QFI = 0.64; the SLD is $L_0 = rZ$.

Both displayed vectors have radius 0.8; their 25° separation is only illustrative.

Changing the measurement changes accessible information, not the underlying quantum limit.

Exact mixed-qubit model; θ in radians. Full rank keeps every displayed measurement regular. No finite-difference QFI approximation is used.

Kraus representations are not unique

Different operator sets can describe exactly the same quantum channel.

$$\mathcal{E}(\rho) = \sum_a K_a \rho K_a^\dagger, \quad \sum_a K_a^\dagger K_a = I$$

Example: complete dephasing of a two-mode state in the chosen basis.

Basis projectors

$$K_0 = |0\rangle\langle 0|, \quad K_1 = |1\rangle\langle 1|$$

Random phase flips

$$\widetilde{K}_0 = \frac{I}{\sqrt{2}}, \quad \widetilde{K}_1 = \frac{Z}{\sqrt{2}}$$

$$\sum_a K_a \rho K_a^\dagger = \frac{\rho + Z\rho Z}{2} = \text{diag}(\rho)$$

Kraus freedom changes the representation. Physical hypotheses concern the channel.

Two-mode illustration of the thesis HG-dephasing model at $v = 0$. Equivalent representations cannot be distinguished by system-only data.

A physical construction: mode-dependent loss

Mode transmission $0 \leq \tau_j \leq 1$; loss is represented by an orthogonal failure state.

Choose operators with explicit physical roles

$$K_t = \sum_j \sqrt{\tau_j} |j\rangle \langle j|, \quad K_j^f = \sqrt{1 - \tau_j} |\text{fail}\rangle \langle j|$$

Transmission and failure jointly account for every input photon.

$$K_t^\dagger K_t + \sum_j (K_j^f)^\dagger K_j^f = \sum_j [\tau_j + (1 - \tau_j)] |j\rangle \langle j| = I$$

Kraus form ensures complete positivity; completeness proves trace preservation.

Channel validity can be proved mathematically; physical suitability requires calibration.

Derived from the thesis mode-dependent loss map. Input: one-photon mode space after common survival; failure is retained in the output.

Information retained by the optical receiver

Compare quantum and classical information at successive stages of the receiver.

$$\mathcal{F}_Q(\rho_S) \geq \mathcal{F}_Q[\mathcal{E}(\rho_S)] \geq \mathcal{F}_C(\mathbf{p}) \geq \mathcal{F}_C(R\mathbf{p})$$

For channels and measurements independent of the parameter being estimated.

Checkpoint

Fraction of source QFI

● Resolved reference, first-order and failure outputs	99.68–100%
● Binary readout; assumed 0.35% port confusion	0.13–57.0%

Aggregation and readout confusion can strongly reduce the accessible information.

Conditional model: same nine settings, per photon surviving common loss. Ranges across settings; not reconstructed experimental QFI.